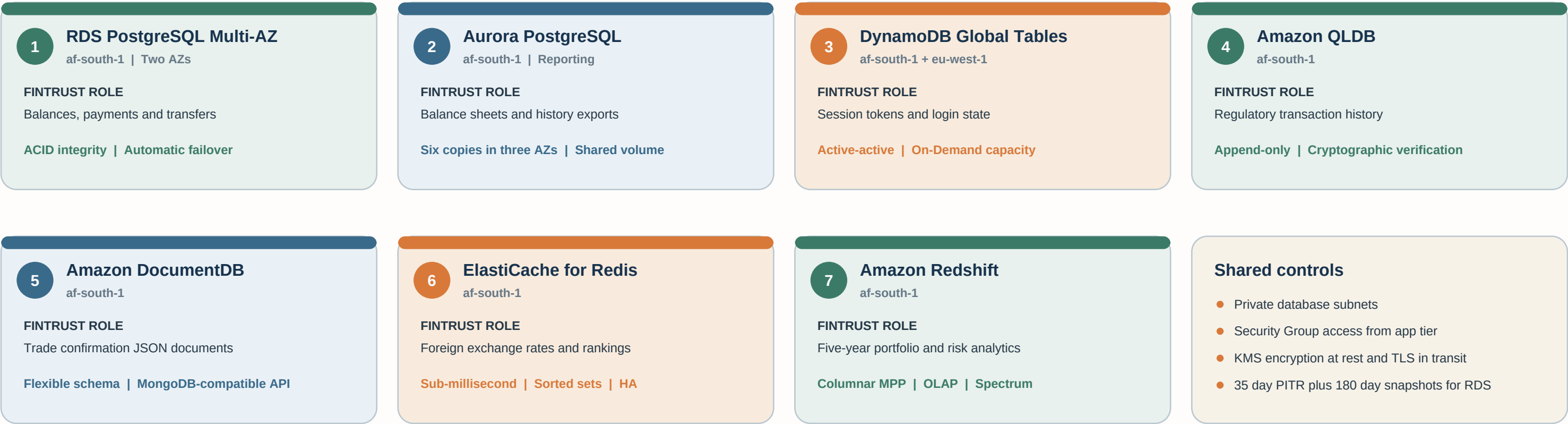


FinTrust seven-layer database architecture

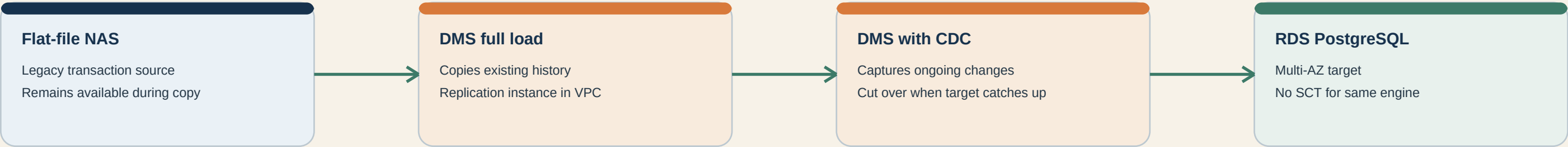
Service map, Region placement and the reason for each database choice

Seven services, seven distinct data patterns

The transaction engine remains relational. Sessions, audit history, documents, caching and analytics use services designed for those access patterns.



Migration path required by the Week 4 brief



Choose by workload

Transactions	RDS PostgreSQL Multi-AZ for relational ACID writes
Reporting	Aurora replica or RDS Read Replica to separate read load
Sessions	DynamoDB Global Tables for active-active Region writes
Audit history	QLDB for an immutable, cryptographically verifiable journal
Trade JSON	DocumentDB for flexible documents and MongoDB-compatible access
Rates cache	Redis for sorted sets, persistence and Multi-AZ failover
Historical BI	Redshift for columnar MPP analytics over large datasets

Redshift decision

Workload	OLAP aggregations, business intelligence and historical trends
Storage	Columnar layout reads only required columns and improves compression
Compute	Massively parallel nodes divide large analytical queries
Spectrum	Queries Parquet, CSV, ORC or JSON in S3 without loading it
RA3	Managed storage separates compute capacity from stored data
Scaling	Concurrency Scaling adds capacity during query peaks
Boundary	Customer transaction writes stay on RDS or Aurora, not Redshift

DMS and SCT

Full load	One-time copy of the data already present at the source
CDC	Replicates inserts, updates and deletes while the source stays live
Production	Full load plus CDC supports a low-downtime cutover
Homogeneous	Same database engine uses DMS without schema conversion
Heterogeneous	SCT converts the schema first, then DMS moves the data
Infrastructure	The DMS replication instance runs inside the VPC

Boundaries that prevent wrong choices

Multi-AZ	Availability and automatic failover, never read scaling
Read Replica	Read scaling through a separate endpoint; promotion is manual
Global Tables	Active-active writes across Regions for DynamoDB
Aurora Global	One writer Region with read-only secondary Regions
DAX	DynamoDB read cache with API compatibility, not a write accelerator
QLDB	Single-authority ledger, distinct from multi-party blockchain
Backups	Automated RDS retention up to 35 days; snapshots cover 180 days